

Employee Wellness Benefit Administration Portal Architecture Document

Isurumuni Wijesooriya
Ashan Odithya
Damath De Silva

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High Level Architecture Diagram

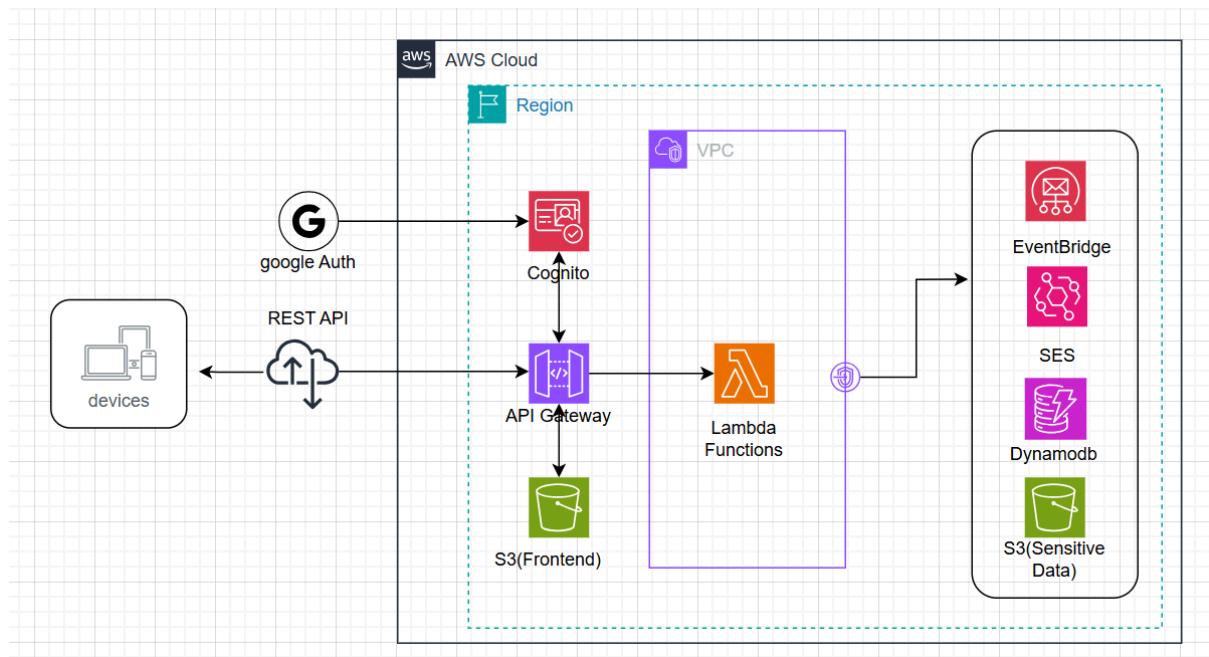


Figure: Deployment Architecture

AWS Services and Usage

SERVICE	DESCRIPTION	USAGE
API Gateway	A fully managed service that makes it easy to create, publish, maintain, monitor, and secure APIs at any scale.	Exposes secure REST APIs for frontend and third-party integrations. Handles request routing, throttling, and authorization before invoking Lambda functions.
DynamoDB	A fully managed, serverless key-value and document database that delivers single-digit millisecond performance at scale, with built-in high availability, durability, and security.	Stores employee profiles, wellness allowance balances, claims, approval status, and transaction history in key-value format.
Google SSO	Google SSO is a centralized	Provides user authentication using

	authentication service that allows users to access multiple applications and services using a single set of Google credentials. It streamlines the login process, enhances security through centralized identity management, and reduces password fatigue for users.	Google identity. Employees and admins authenticate via Google OAuth, enabling secure and centralized identity management.
Lambda	Lambda is an event-driven, serverless computing platform. It is a computing service that runs code in response to events and automatically manages the computing resources required by that code.	Implements backend business logic such as allowance creation, claim submission, approval workflows, eligibility validation, and reimbursement processing.
S3	S3 or Simple Storage Service is a service that provides object storage through a web service interface.	Stores supporting documents such as invoices, receipts, and wellness proof uploads. Also used for audit records and long-term document retention.
EventBridge	EventBridge supports capturing events from multiple sources, applying intelligent routing rules, triggering downstream actions, orchestrating cross-service workflows, and delivering reliable event processing to keep your distributed applications running smoothly.	Manages event-driven workflows such as claim submission events, approval notifications, allowance expiry reminders, and scheduled policy resets.
SES	SES is a cost-effective, flexible, and scalable cloud-based email service designed to send marketing, notification, and transactional emails. It leverages Amazon's high-end infrastructure to ensure reliable delivery and high email deliverability rates.	Sends transactional emails including claim submission confirmations, approval/rejection notifications, allowance expiry alerts, and system communications.

VPC	VPC or Virtual Private Cloud helps to define and launch AWS resources in a logically isolated virtual network.	Provides network isolation and security. Hosts Lambda (via VPC access) and DocumentDB in private subnets, ensuring controlled access and compliance.
Cognito	A fully managed identity and access management service that provides secure user authentication, authorization, and user management for web and mobile applications, with built-in scalability, high availability, and integration with AWS services.	Manages employee user identities, authentication (sign-up/sign-in), role-based access control, and secure access to the application and its APIs, including token-based authentication and integration with IAM for fine-grained permissions.

Architectural Decisions

Service	Choice	Alternative	Why we chose our "Service" over the "Alternative"	Known Limitation
API Management	API Gateway	ALB	Built-in request validation, throttling, and API Key management. ALB is cheaper but requires custom code for security.	Latency & Timeouts: Maximum request timeout is 29 seconds; not suitable for long-running processes.

Compute	Lambda	Fargate	Scales to zero with no idle costs; only pay when a claim is processed. Fargate is wasteful for sporadic claims.	Cold Starts: Occasional latency spikes when a function hasn't been used recently (initial container spin-up).
Database	DynamoDB	DocumentDB	Fully serverless, scales automatically, and provides consistent low-latency with minimal operational overhead.	Complex Queries: Limited support for joins or complex aggregations; requires careful upfront Data Modeling.
Storage	S3	EBS	Object storage via URLs, perfect for web-uploads. EBS is a virtual hard drive limited to one server at a time.	No native file locking or Posix Semantics- Making it unsuitable for shared file usage.
Event Routing	EventBridge	SNS / SQS	Native time-based scheduling (cron) and seamless integration with lower architectural complexity.	Throughput Latency: Higher latency (typically ~500ms) compared to SNS (~tens of ms) for high-frequency events.

Messaging	SES	SNS	Built for rich HTML emails and branded receipts. SNS is for system alerts and lacks deliverability features.	Sandbox Limits: New accounts are restricted to verified emails only until moved to "Production" by AWS Support.
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